T2 TECHNOLOGIES

Feeder Balancing Module (FBM) Tool — Tender Enquiry 3239CXMWP

Author work package — Thabiso

This brief consolidates every section of the tender submission draft assigned to Thabiso, with the current draft content reproduced in full so each section can be developed independently before being merged back into the master document.

Assignment Summary

Section	Title	Ownership
5	Solution Architecture and Design Principles	Owned by Thabiso
6	Delivery Methodology and Implementation Plan	Owned by Thabiso
7	Programme Governance and Quality Management	Owned by Thabiso
17	Commercial Proposal Framework / Pricing Schedule	Shared — Tumelo / Thabiso

5. Solution Architecture and Design Principles

Owned by Thabiso

The target architecture will be confirmed during discovery, but T2 Technologies will apply the following principles unless the Client's standards require otherwise.

Principle	Application
Modularity	Separate business capabilities into clear modules and interfaces so that components can evolve independently.
API First	Expose controlled interfaces for system interaction instead of direct database coupling wherever practicable.
Security by Design	Apply identity, authorisation, encryption, secrets management, auditability and secure coding from the start.
Traceability	Use correlation identifiers and structured logs to trace transactions across service boundaries.
Resilience	Apply timeouts, retries where safe, circuit-breaking or isolation patterns, failure handling and recovery controls.
Idempotency	Protect critical write operations from accidental duplication when business or integration flows can be retried.
Observability	Capture metrics, logs and service health to support proactive operational management.
Data Governance	Define ownership, quality controls, retention, access and lineage for critical data.
Automation	Automate builds, tests, deployments and repeatable infrastructure tasks to reduce operational error.
Portability	Avoid unnecessary technology lock-in and preserve the ability to evolve the deployment model.

5.1 Illustrative Logical Architecture

- Experience Layer: web portals, mobile applications, administration interfaces and assisted-service channels.
- API / Integration Layer: authenticated APIs, routing, orchestration, validation, throttling, transaction controls and external connectors.
- Business Services Layer: domain-specific services, workflows and business rules.
- Data Layer: operational databases, integration stores, reporting data and controlled analytics pipelines.
- Platform Layer: runtime environments, containers/virtual infrastructure, network controls, secrets, backup and recovery.
- Operations Layer: CI/CD, monitoring, logging, alerting, vulnerability management, service management and audit reporting.

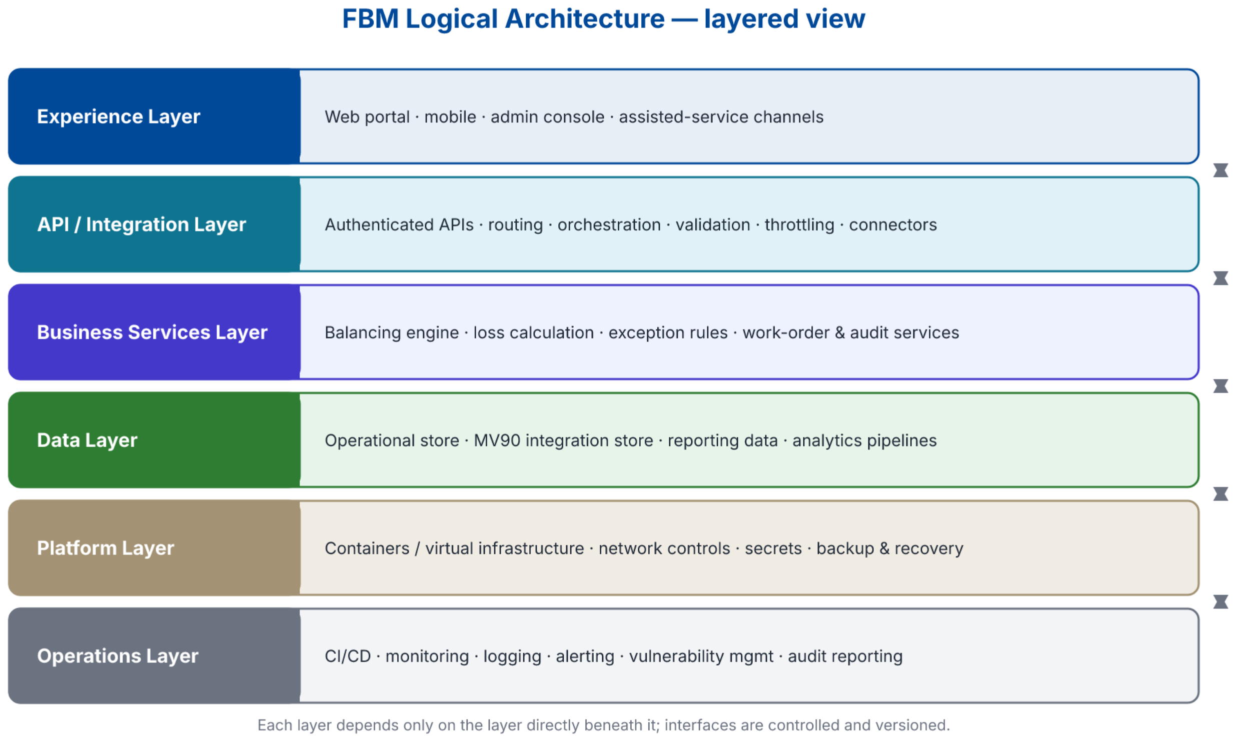


Figure 5.1 — Logical architecture

5.2 Reference Deployment and Data Flow

The module is designed for a resilient, cloud-ready deployment. Traffic is terminated at a managed edge, routed to stateless application services behind a private network boundary, and backed by managed data and integration stores. The data-flow view shows how metered readings move through import, validation and balancing into the decisions and actions the tool surfaces.

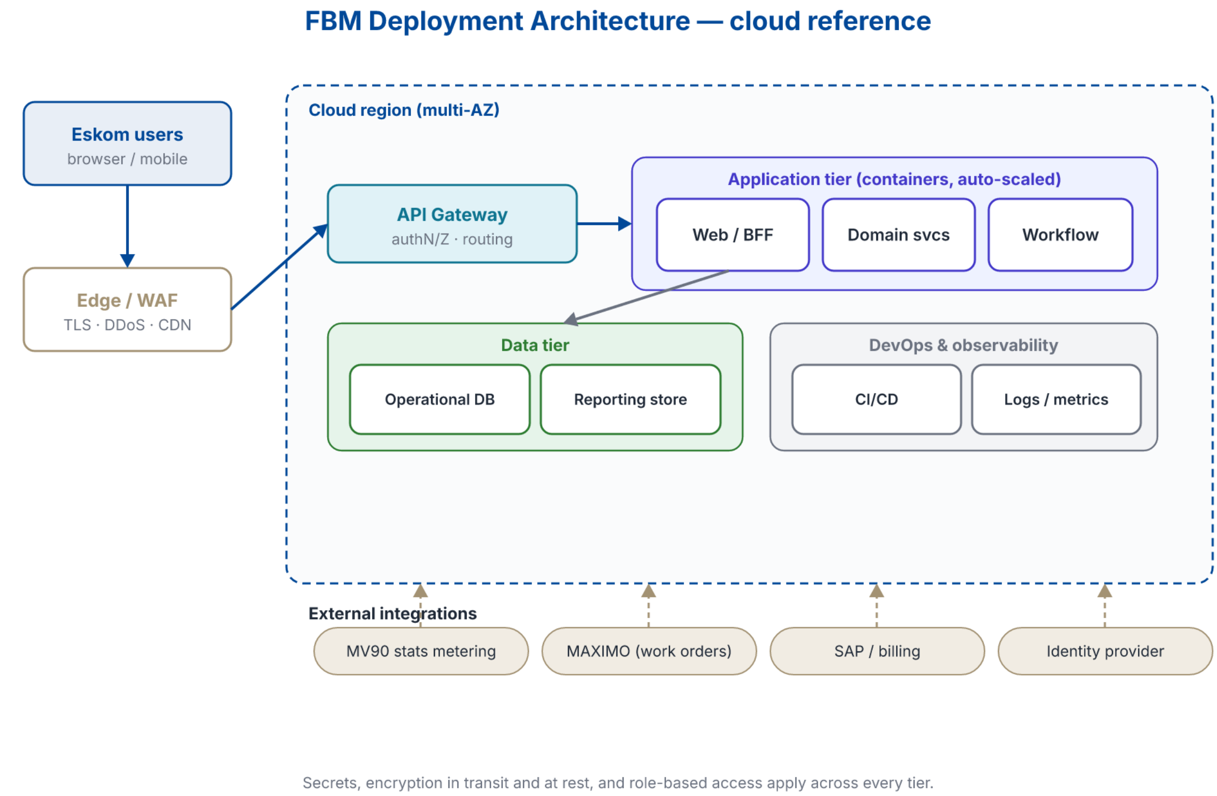


Figure 5.2 — Deployment architecture

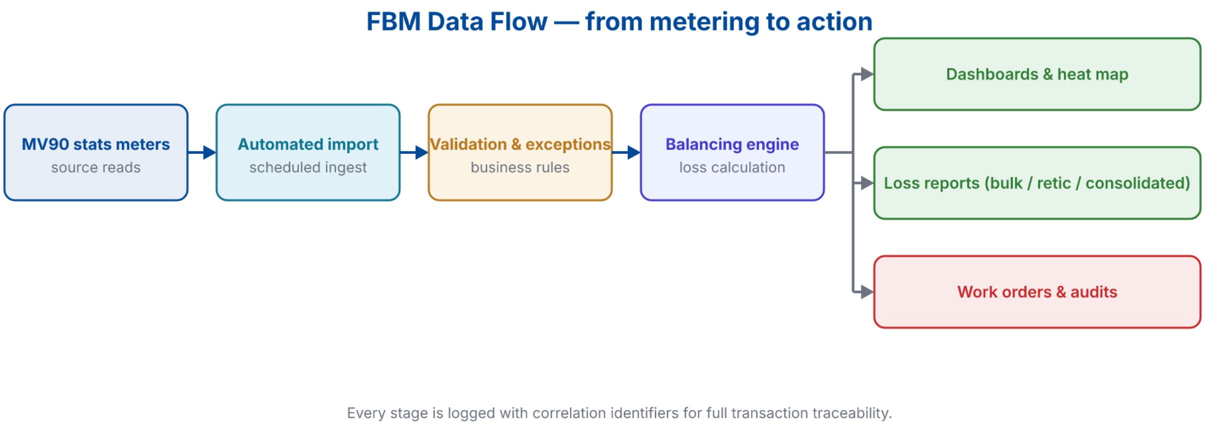


Figure 5.3 — Data flow

5.3 Application Configuration Workflows

The following workflows explain how the tool is operated day to day — behaviour that cannot be conveyed by a static screen alone. Each workflow diagram is paired with a capture of the corresponding screen in the live FBM application.

Configuring a network establishes the eight-level hierarchy (Operating Unit down to Meter); network mapping then lets an engineer re-parent equipment and group meters, transformers or feeders into clusters by drag-and-drop, with energy roll-ups recalculated automatically.

Workflow — Configuring the network hierarchy

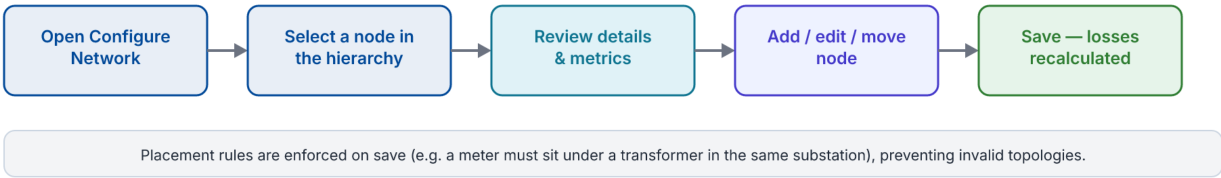


Figure 5.4 — Network configuration workflow

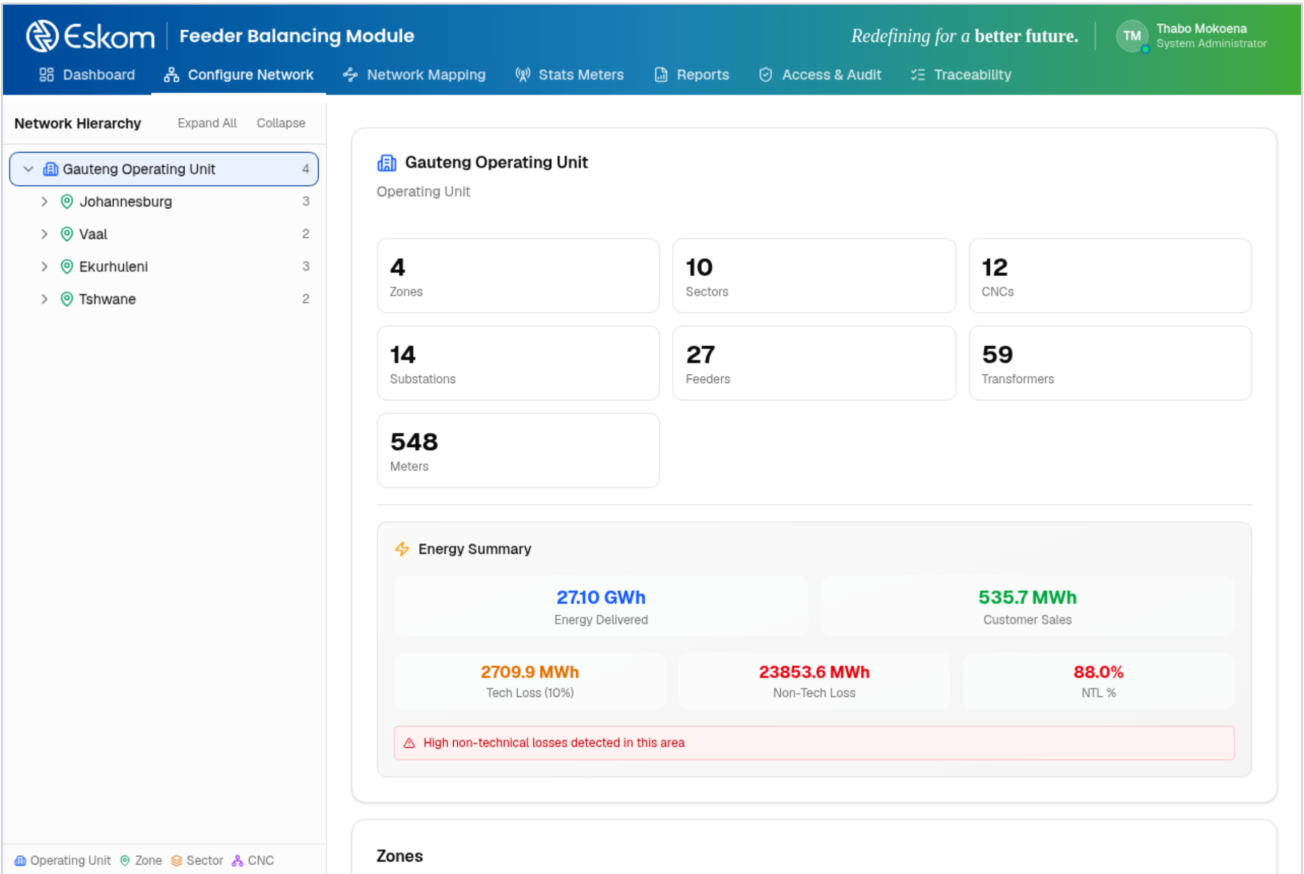
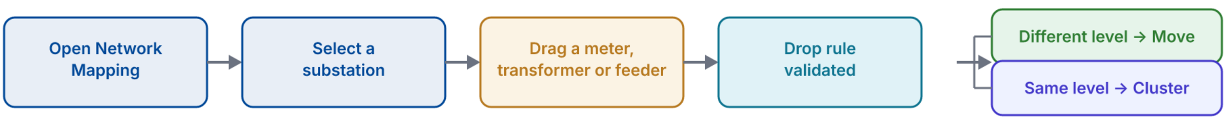


Figure 5.7 — Configure Network (live tool): the eight-level hierarchy with automatic energy roll-up and loss KPIs computed at every level.

Workflow — Network mapping (drag to reorganise)



Drops are validated live; only same-substation, rule-compliant moves are accepted.

Figure 5.5 — Network mapping workflow

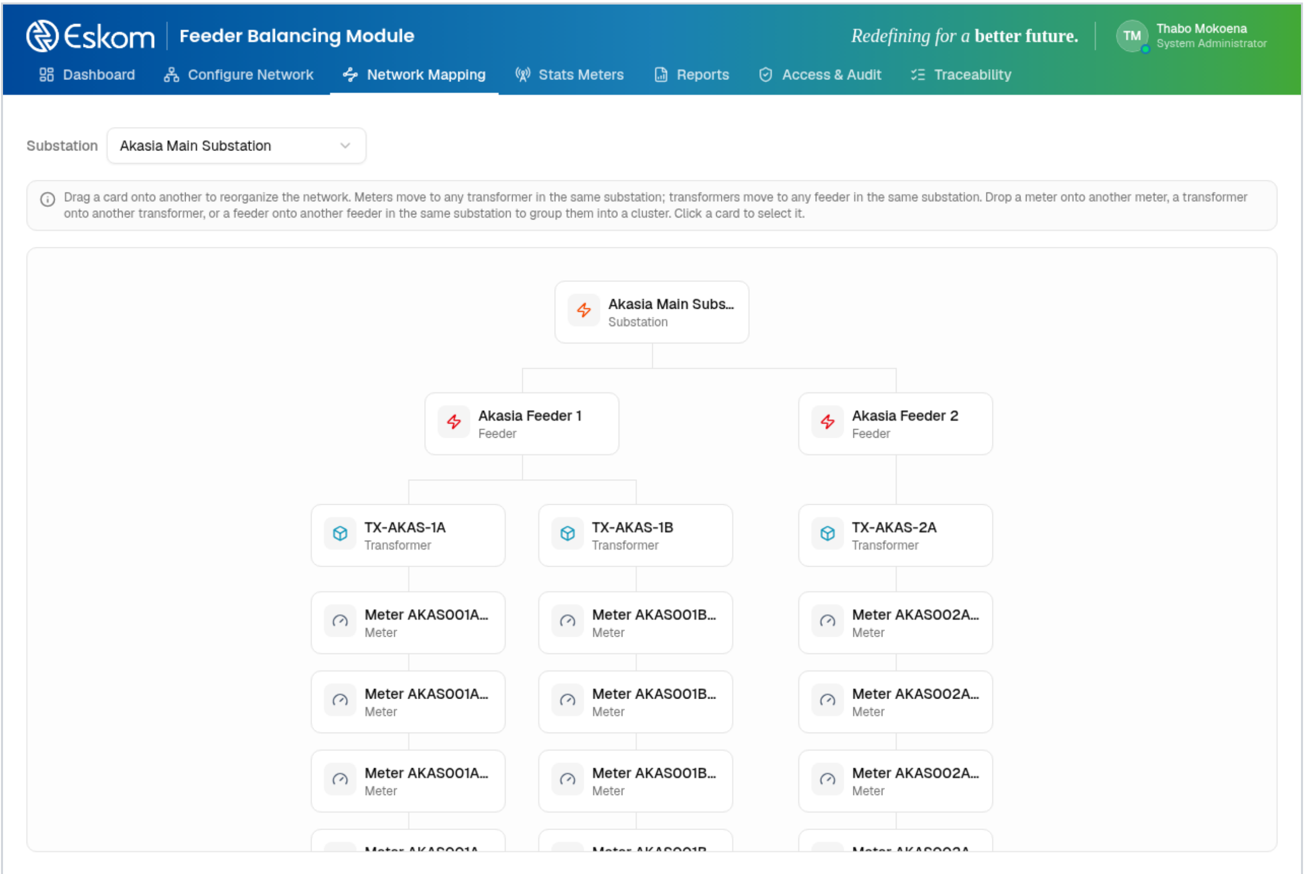


Figure 5.8 — Network Mapping (live tool): drag-and-drop organisation chart for a selected substation.

Workflow — Creating a meter cluster

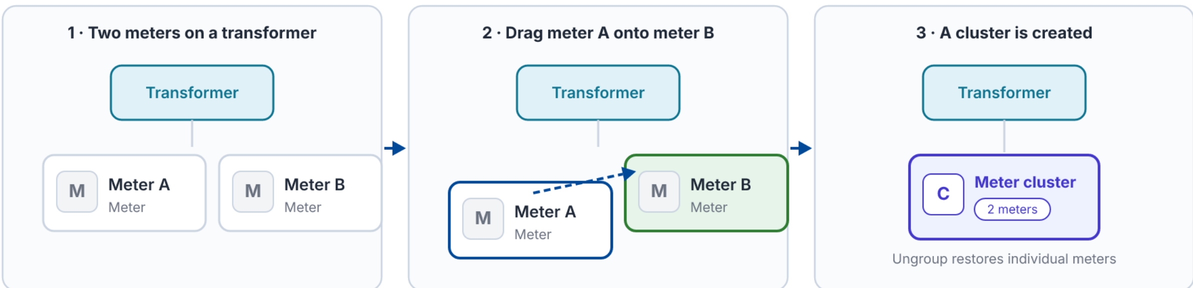


Figure 5.6 — Creating a meter cluster

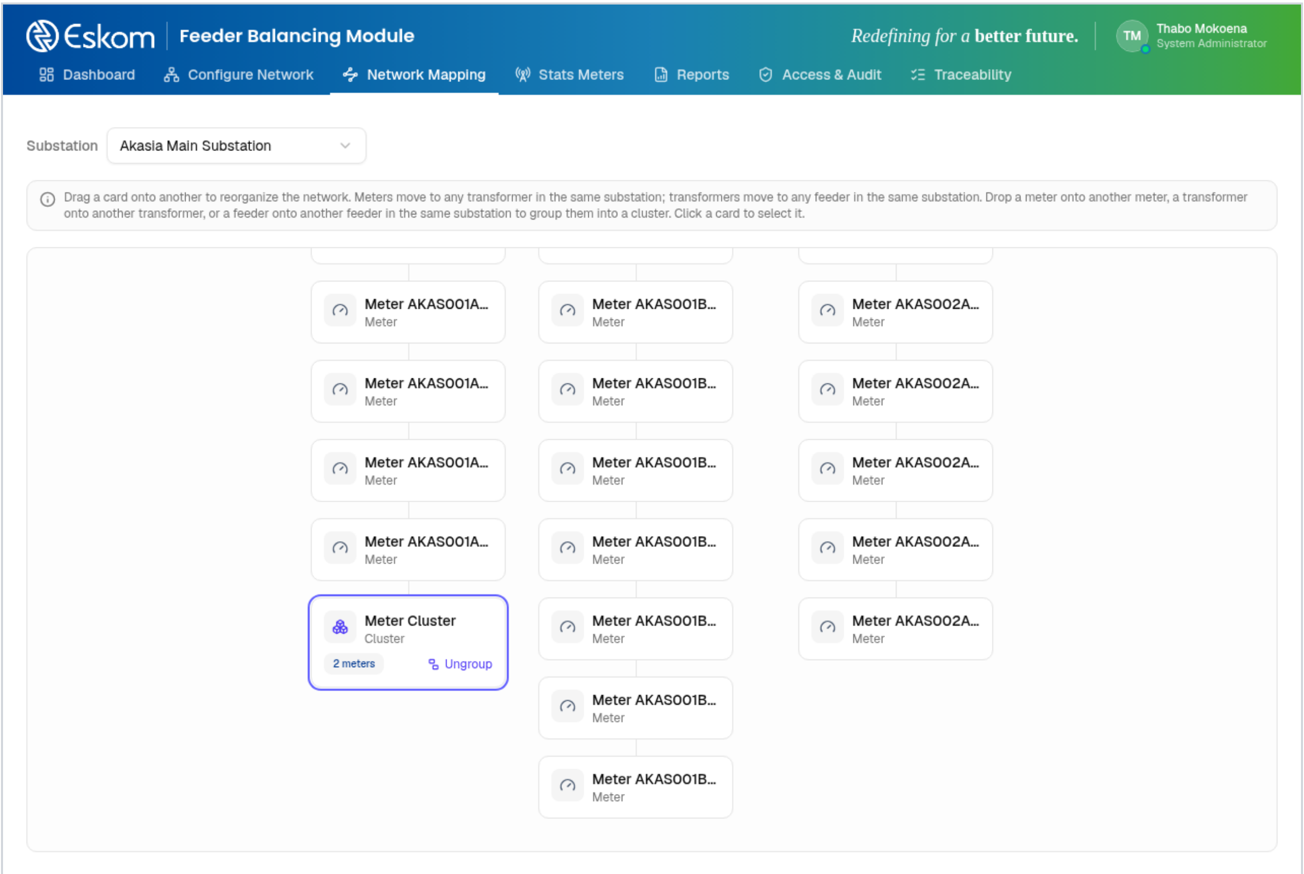


Figure 5.9 — Network Mapping (live tool): two meters grouped into a meter cluster, with one-click Ungroup to reverse the grouping.

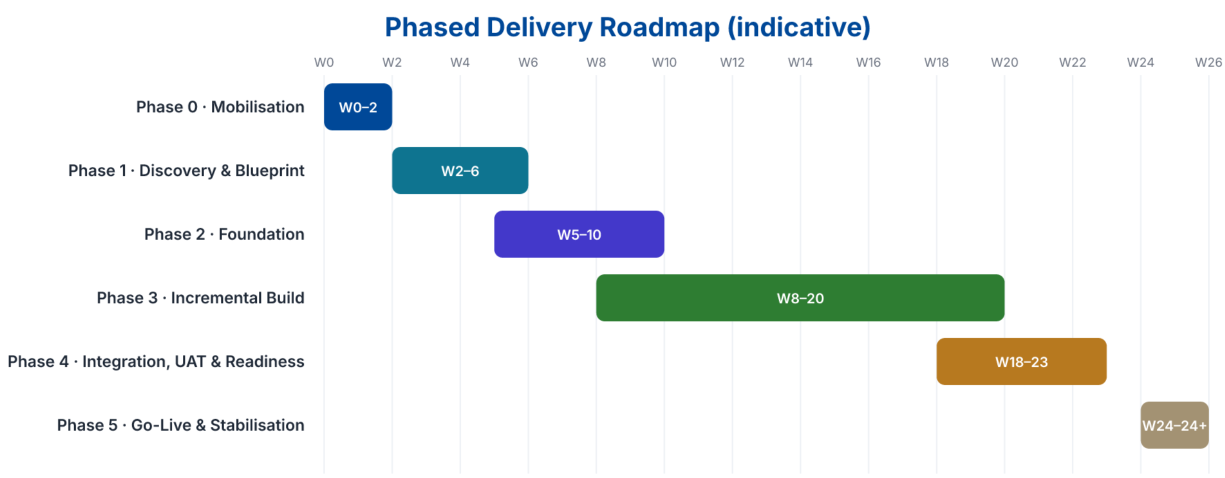
6. Delivery Methodology and Implementation Plan

Owned by Thabiso

T2 Technologies proposes a phased agile delivery model governed by formal stage gates. This balances iterative product development with the traceability and accountability expected in enterprise and public-sector programmes.

Phase	Indicative Timing	Key Activities	Exit Outcome
Phase 0 - Mobilisation	Weeks 1-2	Contract mobilisation, governance, access, stakeholder map, delivery plan, risks, environments and document baseline.	Approved mobilisation pack and detailed project plan.
Phase 1 - Discovery & Blueprint	Weeks 2-6	Current-state assessment, requirements, process mapping, architecture, security assessment and prioritisation.	Requirements baseline, target architecture, backlog and migration approach.
Phase 2 - Foundation	Weeks 5-10	Core platform, CI/CD, security baseline, integration foundation, environments and observability.	Operational technical foundation ready for solution increments.
Phase 3 - Incremental Build	Weeks 8-20	Applications, APIs, integrations, workflows, data pipelines and automated tests delivered in sprints.	Demonstrable functional releases with traceability.
Phase 4 - Integration, UAT & Readiness	Weeks 18-23	End-to-end testing, performance/ security testing, user acceptance, training, migration rehearsal and operational readiness.	UAT sign-off and go-live readiness approval.
Phase 5 - Go-Live & Stabilisation	Weeks 24+	Production deployment, hypercare, defect resolution, monitoring, support transition and post-implementation review.	Stable production service and transition to BAU support.

The above timeline is illustrative. Th
volume, data migration complexity,



Illustrative only — final schedule reconciles to issued tender milestones, scope volume and dependencies.

Figure 6.1 — Delivery roadmap

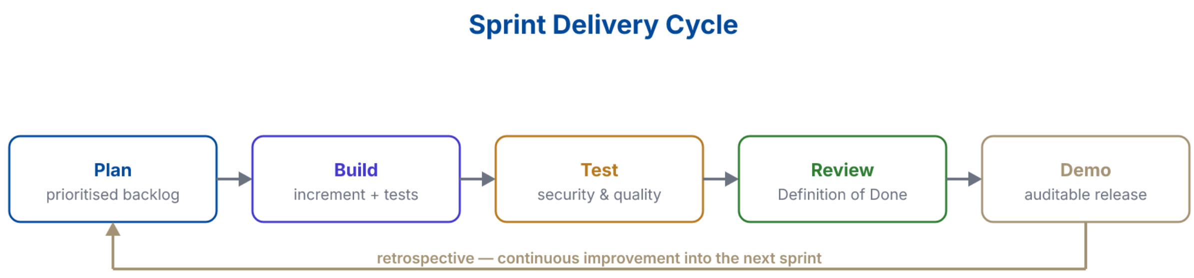


Figure 6.2 — Sprint cycle

6.1 Sprint Delivery

- Prioritised product backlog linked to approved requirements.
- Regular planning, refinement, demonstrations and retrospectives.
- Definition of Done covering functional behaviour, testing, security, documentation and deployability.
- Formal change control for scope or contractual impacts outside the approved backlog baseline.
- Release notes and auditable evidence for each production release.

7. Programme Governance and Quality Management

Owned by Thabiso

Governance will provide clear authority, reporting, escalation and evidence-based decision making. T2 Technologies recommends a joint structure that includes executive oversight, programme management and technical working groups.

Forum	Purpose	Indicative Cadence
Steering Committee	Executive decisions, scope/budget oversight, strategic risks, escalations and milestone approval.	Monthly or as prescribed
Programme Review	Progress, dependencies, RAID log, commercial status, upcoming milestones and actions.	Weekly
Architecture / Security Review	Design decisions, integration standards, security controls, technical debt and exceptions.	Weekly / per major design
Sprint Ceremonies	Planning, stand-ups, demonstrations, retrospectives and backlog refinement.	Per sprint
Service Review	SLA performance, incidents, problems, capacity, releases and continuous improvement.	Monthly post go-live

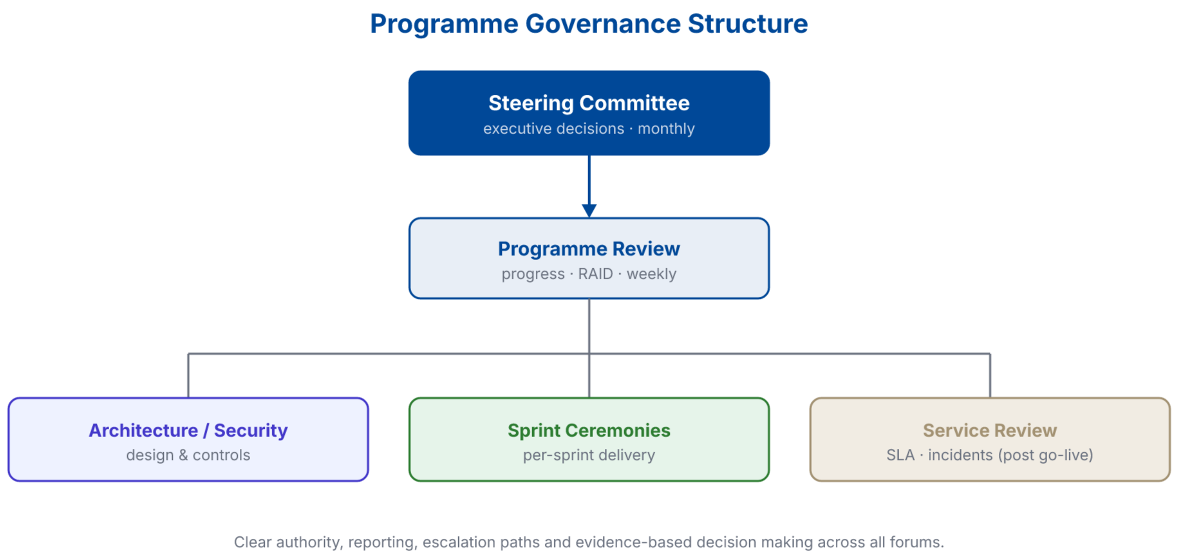


Figure 7.1 — Governance structure

7.1 Quality Controls

- Requirements-to-test traceability.
- Peer review and controlled source-code management.
- Automated unit/integration testing wherever practical.
- Security checks integrated into the development lifecycle.
- Controlled environments and release approvals.
- Document versioning and approval records.
- Defect prioritisation and transparent closure evidence.
- Post-release verification and service health monitoring.

17. Commercial Proposal Framework / Pricing Schedule

Shared — Tumelo / Thabiso

This section presents a fully costed commercial estimate aligned to the structure of the tender's Pricing Schedule (Annexure L): a software/subscription licence line, itemised implementation work-packages, a professional-services rate card, support & maintenance, a five-year total cost of ownership and a milestone-based payment plan. All figures are in South African Rand and exclude VAT unless stated otherwise.

The estimate assumes an 18-month implementation of the business requirements followed by managed support & maintenance to the five-year (60-month) mark, delivered by a standard blended team of approximately 8-10 resources. Every currency figure is derived from the rate card and the effort plan below, so the totals reconcile across all tables.

Commercial validity period, price escalation, retention, performance guarantees and any exchange-rate assumptions must still be completed exactly as required by the tender conditions before the figures are transferred to the official Annexure L pricing forms. All estimates are benchmark-based and subject to T2 Technologies' final review.

17.1 Pricing Basis and Assumptions

- All rates are South African enterprise-IT benchmark day-rates (2026) and must be confirmed against T2 Technologies' actual rate card before submission.
- Prices exclude VAT; VAT at 15% is shown separately in the five-year summary.
- Software is licensed as T2's proprietary SaaS platform on a named-user basis: 20 users at R 3 000 per user per month.
- The implementation is a fixed-price engagement invoiced against the delivery milestones in 17.7; effort is shown for transparency and change-control.
- Support & maintenance begins at production go-live (a three-month warranty is included in Stabilisation) and runs to the five-year mark.
- Third-party cloud/hosting, network connectivity, non-standard integrations, hardware and travel/disbursements outside Gauteng are excluded and quoted at cost on confirmation.

17.2 Software / Subscription Licence

Item	Qty	Unit Basis	Annual (excl. VAT)	5-Year (excl. VAT)
T2 FBM SaaS — named-user licence	20	R 3 000 / user / month	R 720 000	R 3 600 000

17.3 Professional Services Rate Card

The following blended day-

Role	Day Rate (excl. VAT)
Programme / Project Manager	R 9 800
Solution Architect	R 12 500
Business Analyst	R 8 200
Senior Software Engineer	R 9 500
Software Engineer	R 6 800
QA / Test Analyst	R 6 200

Role	Day Rate (excl. VAT)
Data Migration Specialist	R 8 500
DevOps / Cloud Engineer	R 9 200
Change & Training Specialist	R 6 800

17.4 Implementation Work-Package Pricing

Fixed-price implementation broken down

Ref	Work Package	Effort (p-days)	Price (excl. VAT)
WP1	Business Process Analysis	170	R 1 555 000
WP2	Solution Architecture & Design	225	R 2 481 000
WP3	Build, Configure & Test	820	R 6 238 000
WP4	Data Migration	260	R 2 016 000
WP5	Deployment	155	R 1 363 000
WP6	Stabilisation & Go-Live (incl. 3-month warranty)	205	R 1 634 000
WP7	Training & Knowledge Transfer	140	R 994 000
WP8	Change Management	170	R 1 272 000
WP9	Project & Programme Management	320	R 3 190 000
	Total implementation (fixed price)	2465	R 20 743 000

17.5 Support and Maintenance

Item	Basis	Annual (excl. VAT)	Term	Total (excl. VAT)
Managed support & maintenance	Blended support team	R 1 948 100	4 yrs (post go-live)	R 7 792 400

Support covers L2/L3 application s management under an agreed SLA

17.6 Five-Year Total Cost of Ownership

Year	Implementation	Licence	Support & Maint.	Annual Total
Year 1	R 20 743 000	R 720 000	—	R 21 463 000
Year 2	—	R 720 000	R 1 948 100	R 2 668 100
Year 3	—	R 720 000	R 1 948 100	R 2 668 100
Year 4	—	R 720 000	R 1 948 100	R 2 668 100
Year 5	—	R 720 000	R 1 948 100	R 2 668 100

Year	Implementation	Licence	Support & Maint.	Annual Total
Total (excl. VAT)				R 32 135 400
VAT @ 15%				R 4 820 310
Total (incl. VAT)				R 36 955 710

17.7 Payment Milestone Schedule

Implementation value is invoiced
month term.

Milestone	% of Implementation	Value (excl. VAT)
Contract signature & mobilisation	10%	R 2 074 300
Business Process Analysis sign-off	10%	R 2 074 300
Solution design approval	15%	R 3 111 450
Build, configure & test complete	25%	R 5 185 750
Data migration & UAT sign-off	15%	R 3 111 450
Production go-live	15%	R 3 111 450
Stabilisation & final acceptance	10%	R 2 074 300
Total	100%	R 20 743 000